

ORIENTATION

The AI Revolution *Starts Here*

A beginner-friendly guide with real-world analogies.

No tech background needed — just curiosity.



Based on the Agent Factory Thesis
agentfactory.panaversity.org

What We'll Cover Today

01 What is AI, Really?

02 What is Agentic AI?

03 Two Ways to Use AI
(General Agents)

04 The 10-80-10 Rhythm

05 From 'Smart Tool' to 'Smart
Worker'

06 The Agent Factory Idea

07 Your Personal AI Delegate

08 The 7 Principles of General Agent
Problem Solving

09 The 7 Golden Rules of
an AI-Native Company (Invariants)

10 A Real Example: Ecomm Mart

PART 01

What is AI, Really?



ORDINARY MICROWAVE

Uses fixed time & power

SMART MICROWAVE

Looks, recognizes, decides

**ANALOGY**

The Smart Kitchen Appliance

A normal microwave follows fixed rules.

A smart one can look, recognize, and decide what to do.

1**Input food**

You place food inside

**2****Reads what it sees**

Figures out what the food is

**3****Chooses the best settings**

Picks time & temperature automatically



Working with AI is the same

**Information in**

AI looks at the input

**Pattern recognition**

It figures out what it is

**Decision**

It chooses what to do

**AI doesn't follow every step from you. It figures things out from the information it sees.**

Breaking Down "Artificial Intelligence"

Artificial

Made by humans, not by nature.

A plastic flower is artificial.

AI is intelligence built from software.

Intelligence

The ability to learn, understand,
and make decisions.

Humans have natural intelligence.

AI has a version built from data.

Everyday Examples You Already Use

Google Translate

Reads Urdu, gives
English

YouTube Recs

Spots patterns in what
you like

Auto-correct

Predicts the word you
meant

ChatGPT / Claude

Writes answers from
questions

KEY TAKEAWAY



AI is software that can learn from data and make decisions on its own — without a human giving it step-by-step instructions every time.

It's not magic. It's math, data, and clever programming.

And it's already everywhere in your daily life.

PART 02

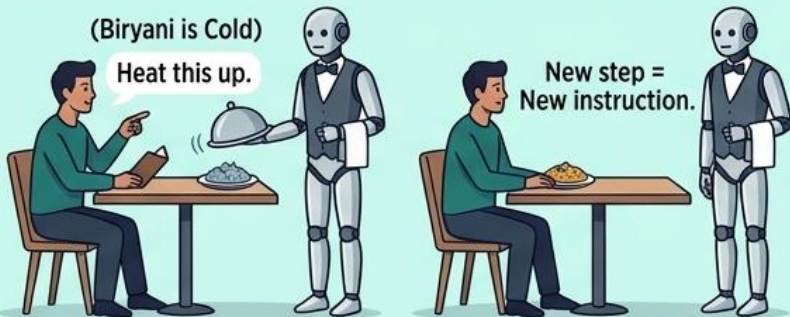
What is Agentic AI?



AI EXPLAINED: THE RESTAURANT ANALOGY

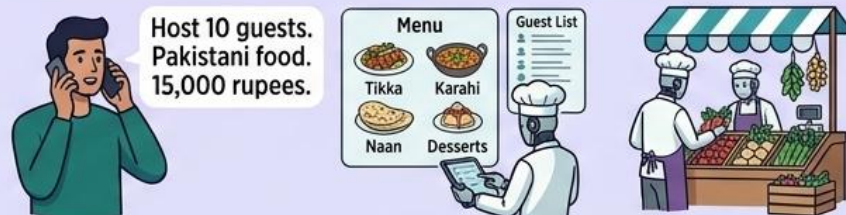


REGULAR AI: THE WAITER (REACTIVE)



- Explicit Orders Only
- New Step, New Command
- No Proactivity

AGENTIC AI: THE PERSONAL CHEF (PROACTIVE)



- Goal-Oriented
- Self-Directing
- End-to-End Execution
- No Hand-Holding

Three Levels of AI

LEVEL 1

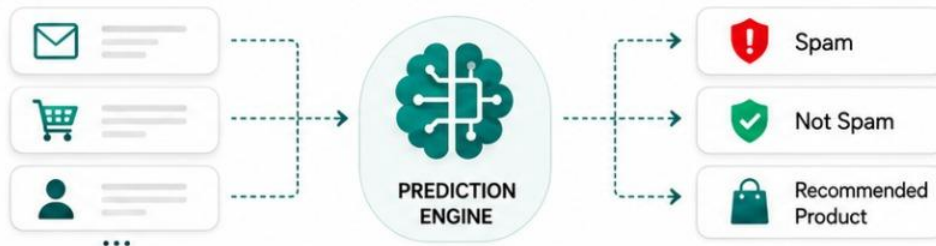


Level 1 — Predictive AI

(The Smart Filter)

You give it data. It gives one prediction.

Spam or not spam? What product to recommend? No conversation, no memory.



LEVEL 2

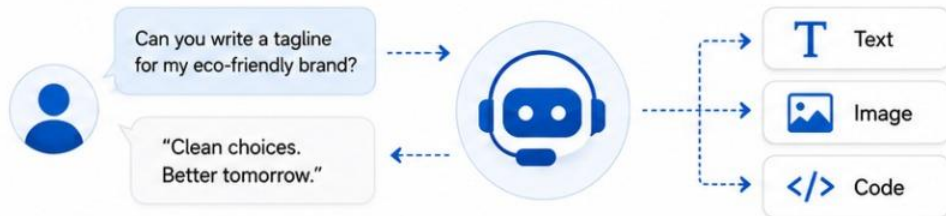


Level 2 — Generative AI

(The Chat Partner)

You have a conversation. It remembers context.

Creates text, images, and code.
Still waits for your next instruction.



LEVEL 3



Level 3 — Agentic AI

(The Worker)

You give it a goal. It does the work.

It plans, uses tools, fixes mistakes, and finishes the job — without you guiding every step.



Level 1
predicts.



Level 2
creates and chats.



Level 3
plans and delivers.



What Makes AI "Agentic"?

Four abilities that turn a chatbot into a worker:



Planning

Breaks a big goal into smaller steps and decides the order



Tool Use

Browses the web, writes code, reads files, sends messages



Self-Correction

Notices mistakes and tries a different approach on its own



Goal Completion

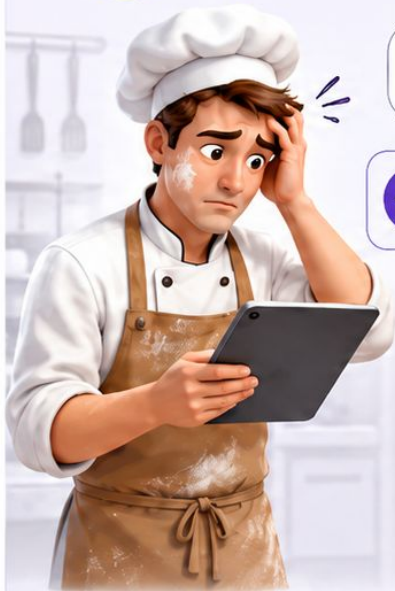
Works until the job is done, not just one answer



Bakery Email Marketing: Two Paths



Regular AI: The Single-Task Tool



"Write a marketing email for my bakery."



Writes **ONE** generic email.
Then it is done.

Your manual tasks:



1 Send the email manually



2 Track opens and clicks



3 Follow up with interested customers



Agentic AI: Your Intelligent Partner

"Run my email marketing this month."



Manages the **entire** campaign.



1. Segments customers



2. Writes targeted personalized emails



3. Schedules automatic sending



4. Tracks performance and improves the next batch



Regular AI helps with one task. **Agentic AI** handles the full goal.

Tool = one step | Agent = end-to-end work



AI Personal Agents vs General Agents

Two different AI roles in the age of AI-native computing

AI Personal Agents



Hi! I'm here
to help you.



OpenClaw



Also: **Hermes**

- Works for one person
- Knows your context, preferences, and goals
- Helps with everyday tasks and personal workflows
- Stays close to your files, apps, and routines
- Acts like a proactive personal AI companion



Best for



Personal
productivity



Reminders



Planning



Research
assistance



Life
organization

VS

General Agents



Claude Code

Developer-focused
agent for building
and engineering



Claude Cowork

Collaborative agent
for teams and
organizations

- Works inside a broader environment
- Uses tools, files, and workflows to complete real tasks
- Can support developers and domain experts
- More task-oriented than personally intimate
- Designed for work across projects and organizations



Best for



Coding



Building



Testing



Research



Analysis



Writing



Coordination



Key Difference

Personal agents are centered on one individual and their daily life.
General agents are centered on getting broader work done across tools, projects, and professional environments.

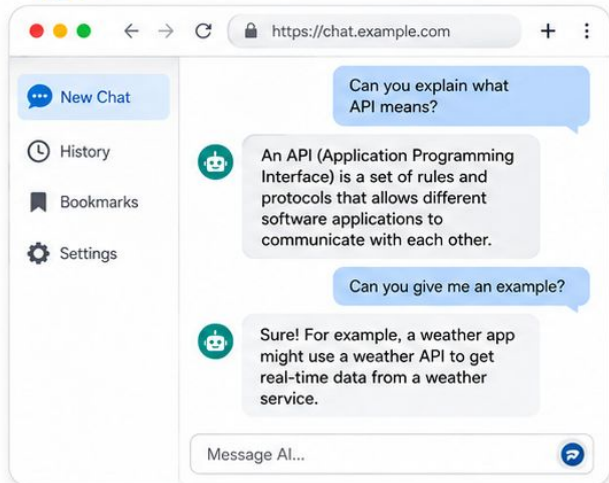


Example framing: **OpenClaw** = personal agent | **Claude Code** and **Claude Cowork** = general agents

From Browser Chat to General Agents

How most readers first meet AI — and what comes next

1. Browser Chat



Most readers first meet AI through a browser-based LLM. They log into ChatGPT, Claude, Gemini, or another model and start typing in a chat window.

ChatGPT

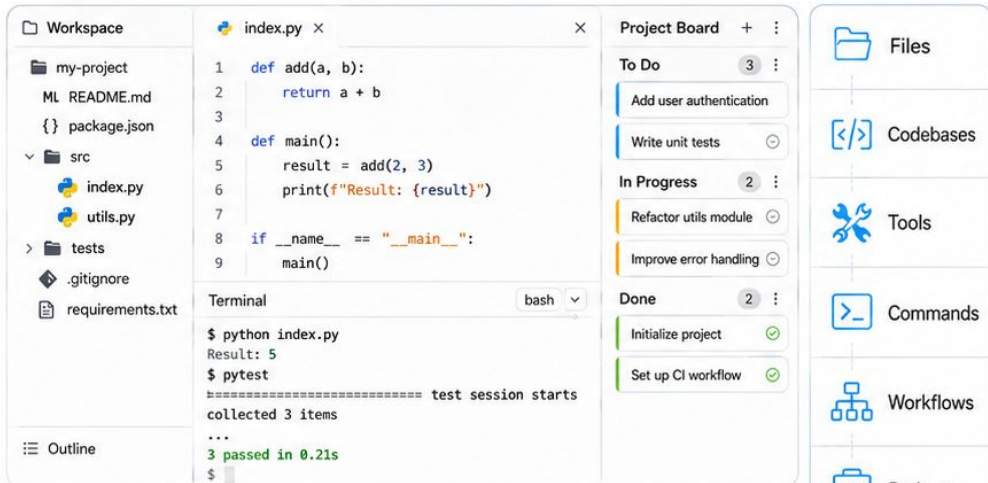
Claude

Gemini

Other LLMs

Move
Beyond Chat

2. General Agents



General agents move beyond ordinary chat. They can work inside an environment: files, codebases, tools, commands, workflows, and projects.

Claude Code

OpenCode

Claude Cowork

OpenWork



Chat: conversation in a browser

Ask questions. Get answers. Stay in the chat.



General agents: work inside an environment

Take actions. Use tools. Complete real work.



Old model:

humans use apps directly

AI-native model:

humans delegate work
through general agents

General Agents as the AI Operating Layer

Not the operating system itself – but the layer that connects human intent to real work

4

Work Environment



Files



Codebases



Documents



Apps



Tools



Commands



Workflows



Projects

3

AI Operating Layer



For Developers

Claude Code

OpenCode

Build, test, refactor, deploy

For Knowledge Work

Claude Cowork

OpenWork

Research, analyze, write, coordinate



2

Human Interface Layer



Developers

IDE • Terminal • Browser • File System



General Users & Domain Experts

Browser • Docs • Email • Spreadsheets • SaaS Apps

1

Traditional Operating System



Windows • macOS • Linux

Manages files, memory, processes, devices, and security



Browser chat:

ask questions and
stay in the chat

.....

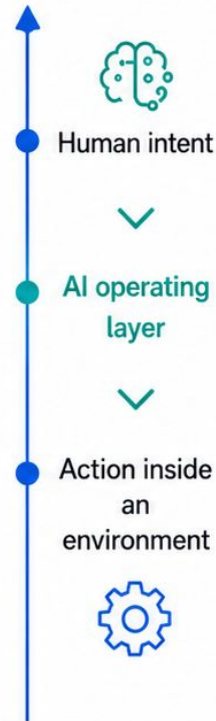
VS

.....

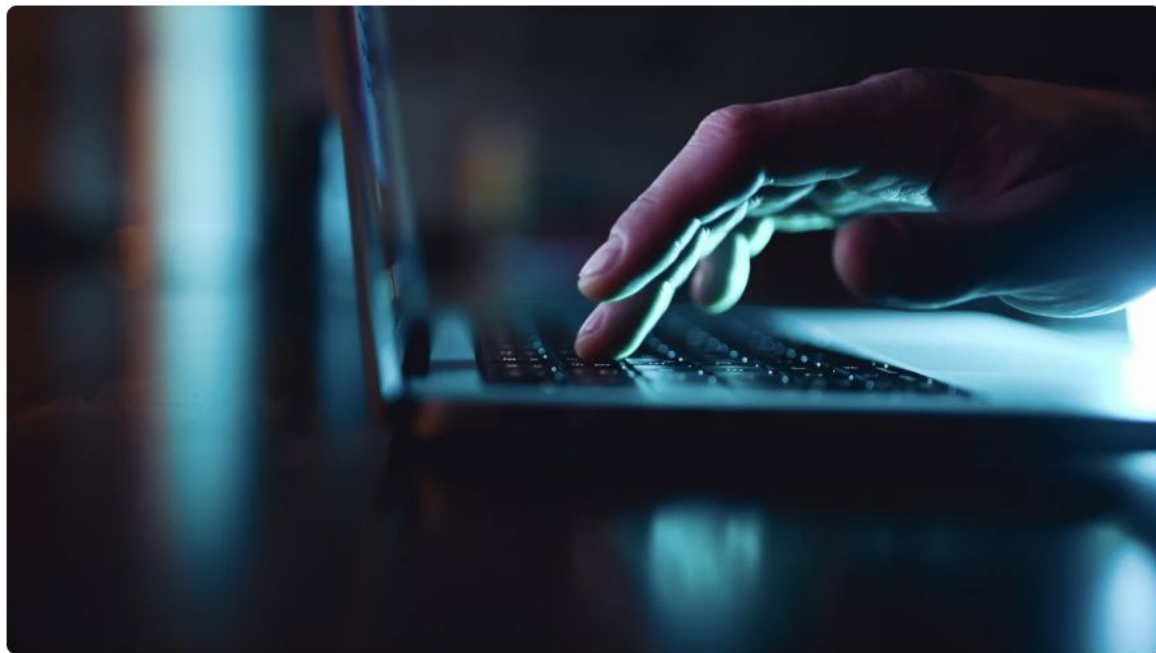


General agents:

work inside an environment
and complete real tasks



BUSINESS & TECH



The world's biggest tech companies are betting big on computers that control themselves



NVIDIA RTX SPARK

A POWERFUL NEW CHAPTER
FOR WINDOWS PCs

BUILT FOR THE AGE OF AI



UP TO
1 PETAFLOP
AI PERFORMANCE



UP TO
20 ARM
CPU CORES



UP TO
6,144
RTX CORES



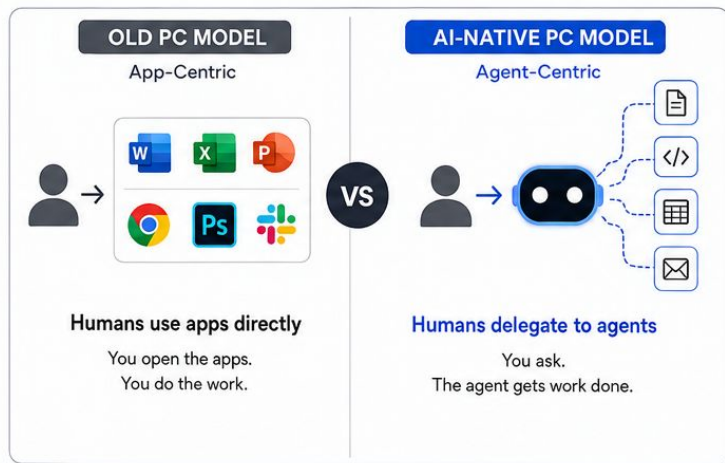
UP TO
128GB
UNIFIED MEMORY



INDUSTRY-LEADING
PERFORMANCE
PER WATT

THE NEW ERA OF COMPUTING: BUILT FOR AGENTS

From App-Centric PCs to Agent-Native PCs



A NEW CLASS OF PC: THE PERSONAL AGENT PC
Purpose-built for personal agents and general agents

Example: Microsoft Surface Laptop Ultra
Powered by NVIDIA RTX Spark

- Up to 1 petaflop of AI performance
- Up to 128GB unified memory

Built for Windows
Optimized for a new era of AI experiences

Runs locally, under your control
Your data stays with you.
Private, secure, and powerful.

BUILT TO RUN PERSONAL AGENTS AND GENERAL AGENTS

Understand

Reason

Plan

Act

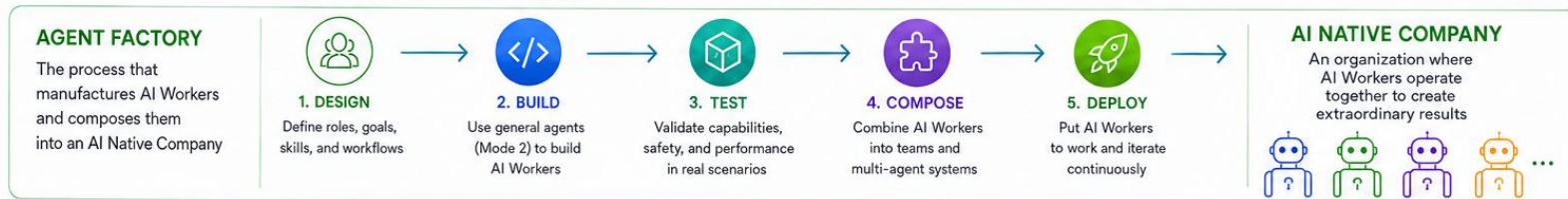
Use Your Apps

Access Your Files

Execute Workflows

DEVELOPER MODE (Mode 2)
We use general agents (Claude Code, OpenCode) to manufacture AI Workers

OUTPUT: AI WORKERS
Specialized AI agents that can think, use tools, and complete real work



PERSONAL AGENTS
Always with you.
Working for you.

OpenClaw
Your personal AI assistant.
Knows you. Works for you.
Helps across all your tasks.

Hermes
Your proactive AI companion.
Plans ahead. Takes action.
Gets things done for you.



THE FUTURE:

People set direction. Agents do the work. Companies scale intelligence.

Built for the Age of AI

NVIDIA RTX Spark Laptop vs MacBook Pro with M5 Max

Two Arm-based laptops. Built for AI. Designed for the future.



RTX Spark Laptop

Windows AI Laptop
Powered by NVIDIA RTX Spark

- Up to 1 petaflop AI performance
- Up to 128GB unified memory
- Windows 11 AI PC
- Built for personal & general agents



Arm CPU

Up to 20 cores
(Performance + Efficiency)

NVIDIA Blackwell
GPU with Tensor Cores



MacBook Pro with M5 Max

macOS

- 16-core Neural Engine for on-device AI
- Up to 128GB unified memory
- macOS Optimized for on-device AI
- Built for creators, developers and professionals



Arm CPU

Up to 16 cores
(Performance + Efficiency)

Up to 40-core GPU

VS

KEY DIFFERENCES

- Higher AI Performance**
Up to 1 petaflop of AI performance with NVIDIA RTX Spark.
- More Memory for AI**
Up to 128GB unified memory optimized for AI workloads.
- Windows AI Ecosystem**
Access to a broad range of AI apps, tools, and enterprise workflows.
- NVIDIA Software Stack**
CUDA, TensorRT, AI libraries, and enterprise-grade support.

NVIDIA RTX Spark Laptop

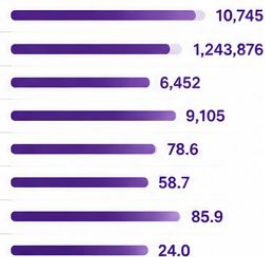


BENCHMARK COMPARISON

Higher is better

- Geekbench 6 (Multi-Core)**
Overall CPU performance
- Geekbench AI (Quantized Score)**
On-device AI performance
- UL Procyon AI Computer Vision**
Image generation (SD 1.5)
- UL Procyon AI Text Generation**
LLM inference (Phi-3.5-Mini)
- Stable Diffusion XL (fp16)**
Images per minute (faster is better)
- Token Generation Speed (fp16)**
Tokens per second (Llama 3.8B)
- Blender 4.2.0 (CPU)**
Classroom (samples per minute)
- Battery Life (Video Playback)**
Hours (longer is better)

MacBook Pro with M5 Max



KEY STRENGTHS

- macOS Optimization**
Deeply integrated software and hardware for efficiency.
- Industry-Leading Battery Life**
Up to 24+ hours of video playback for all-day productivity.
- Privacy by Design**
On-device AI with advanced privacy and security protections.
- Premium Build & Display**
Exceptional build quality and stunning Liquid Retina XDR display.

BOTH ARE BUILT FOR THE AGE OF AI



Run Powerful AI Locally

Execute large models, generate content, and automate tasks on-device.



Private & Secure

Your data stays with you.
AI on your device, on your terms.



Built for Developers

Tools, frameworks, and SDKs to build the next generation of AI experiences.



The Future is Agentic

From personal assistants to autonomous agents—both are ready.

PART 03

Two Ways to Use AI (General Agents)



Our main AI agents are general agents

General agents adapt across many tasks instead of being limited to one narrow workflow.

For Developers



Claude Code OpenCode



General agents for developers

Used for coding, building, debugging,
and software problem-solving.

For Domain Specialists



Claude Cowork OpenWork



General agents for domain
specialists solving problems

Used for analysis, research, planning, writing,
and business-domain problem-solving.



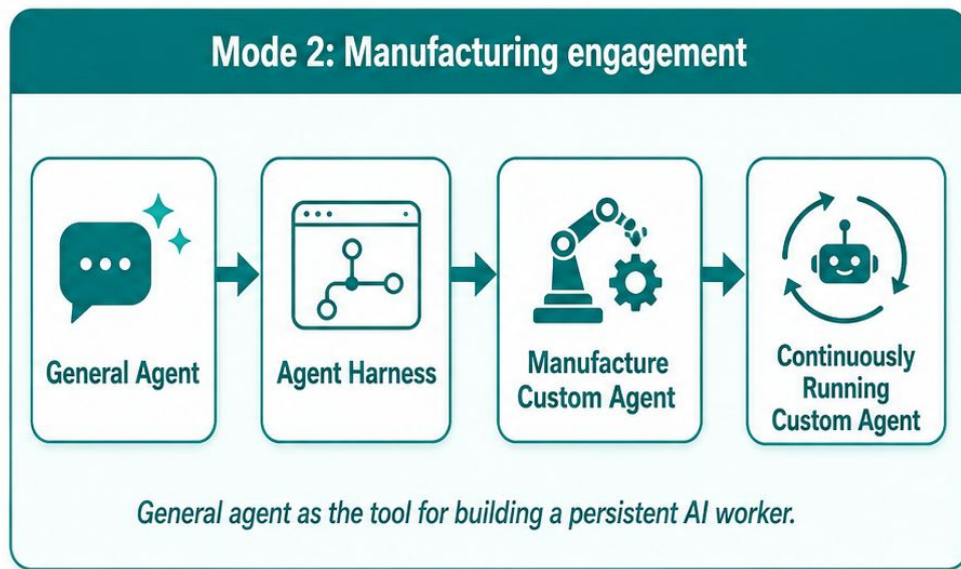
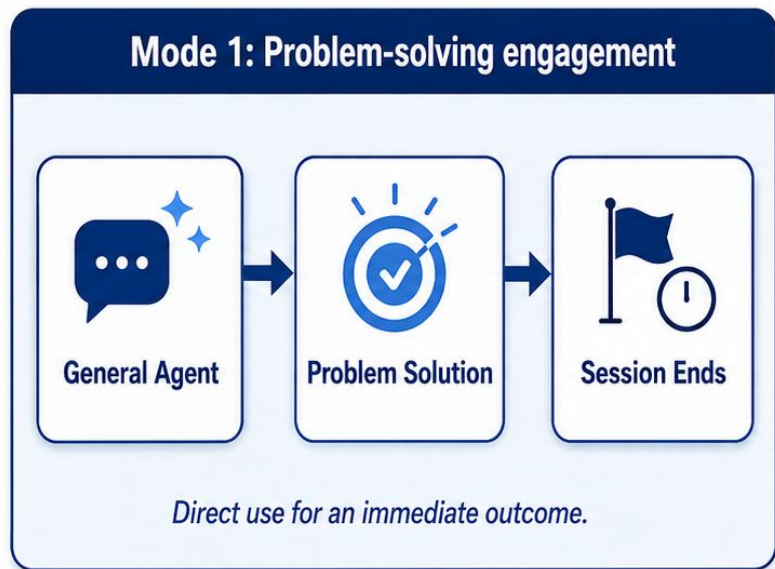
⇒ Both are general agents ⇒

Different audiences, same core idea: flexible agents that help people solve real work problems.

Two Modes of General Agent Use

Mode 1 uses a general agent to solve an immediate problem.

Mode 2 uses a general agent to help build and deploy a custom agent that can keep running.

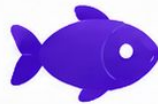


Refinement: in Mode 2, the continuously running custom agent is typically deployed through an agent harness or runtime, rather than the general agent itself running continuously.



ANALOGY

Fish vs. Fishing Net



Two Modes. Two Outcomes.

MODE 1

PROBLEM-SOLVING

Catch a fish. Solve today's problem.



CATCH ONE!



EAT IT



PROBLEM
SOLVED



Solves today.



Start with Mode 1
Solve today's problem.



BOTH ARE
VALUABLE



MODE 2

MANUFACTURING

Build a net. Catch fish every day.



BUILD
A NET



CATCH FISH
EVERY DAY



SCALE & GROW



SUSTAINABLE
RESULTS



Builds tomorrow.



Move to Mode 2
Build tomorrow's solution.



Learn both. Use both. Grow beyond both.

Two Modes Side by Side

Mode 1 – Problem-Solving

"Help me do this thing right now."

You have a problem. You use AI to solve it. The session ends. Nothing permanent is built.

Example: "Help me analyze this spreadsheet" or "Write me an email to my client."

Mode 2 – Manufacturing

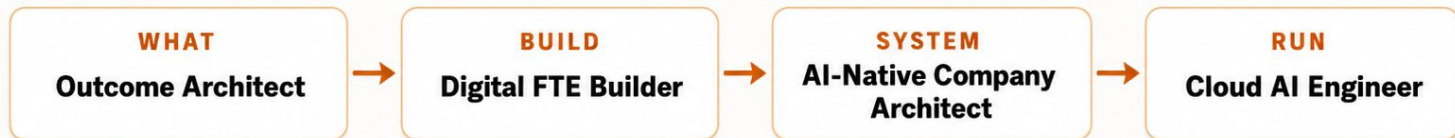
"Help me build a worker that does this job forever."

You use AI to build a new AI worker that will keep running long after you close your laptop.

Example: "Build me a customer support bot that runs 24/7 and answers common questions."

The Roles This Book Trains

1. THE CORE PIPELINE • INTENT → PRODUCTION



2. EXTENDS IT • THE TWO ROLES ONLY THIS BOOK TRAINS

Subject Matter Expert as Skill Author

encodes domain judgment — the Worker's knowledge engine

Forward Deployed Engineer

the whole pipeline, run inside a client's company

3. SUPPORTS IT

Evals Engineer

the verification standard

AI Governance Officer

authors the workforce's rules

Digital FTE Supervisor

accountable, Worker by Worker

4. WHERE THE BOOK STOPS

LLMOps Engineer

ops & fine-tuning, not models

Harness Engineer

uses runtimes, doesn't build them

AI Data Engineer

agent-facing data only

5. THE BASELINE EVERYONE STARTS FROM

Mode 1 Practitioner

use a general agent to do your own work faster — a proficiency, not a title

PART 04

The 10-80-10 Rhythm





— ANALOGY —

The Film Director

A director doesn't do every task.
They **guide** the work.

1



Set the vision

What should
it look like?

2



Let the team work

They do the work

3



Review the final cut

Check and approve



Working with AI is the same



First 10%

You set direction



Middle 80%

AI does the work



Last 10%

You review & approve



START



IN PROGRESS



FINISH

The 10-80-10 Split

10%

Human sets
direction

80%

AI does the work

10%

Human checks
& approves

Three Things That Always Stay With Humans



Intent

Knowing what you want



Verification

Checking if the work is
good



Ownership

Being responsible for the
result

You're Promoted, Not Replaced



The Human Role

You are promoted, not replaced. From typist to editor. From coder to architect. From worker to director. AI takes over the middle 80%. You keep the parts that matter most — direction and judgment.

REAL EXAMPLE

Old Way

You spend 4 hours writing a business report yourself.

New Way

20 min writing clear instructions. AI writes the report in 5 min. 20 min reviewing and fixing. Same quality. One-tenth the time.

PART 05

From "Smart Tool" to "Smart Worker"



OLD WAY

You hold it.
You decide.
You do the work.

AI as a **TOOL**



You control every step



Tool makes one part faster



You still do the job



ANALOGY

The Carpenter's Drill vs. Hiring a Contractor

NEW WAY

You give the job.
It does the work.
You check the result.

AI as a **WORKER**



You define the outcome



Worker handles the task



You review the result



Working with AI is the same



Give the job

Define what you want



AI does the work

It handles the task



Check the result

Review and approve



Tool = You do the work



Worker = AI does the work

KEY TAKEAWAY



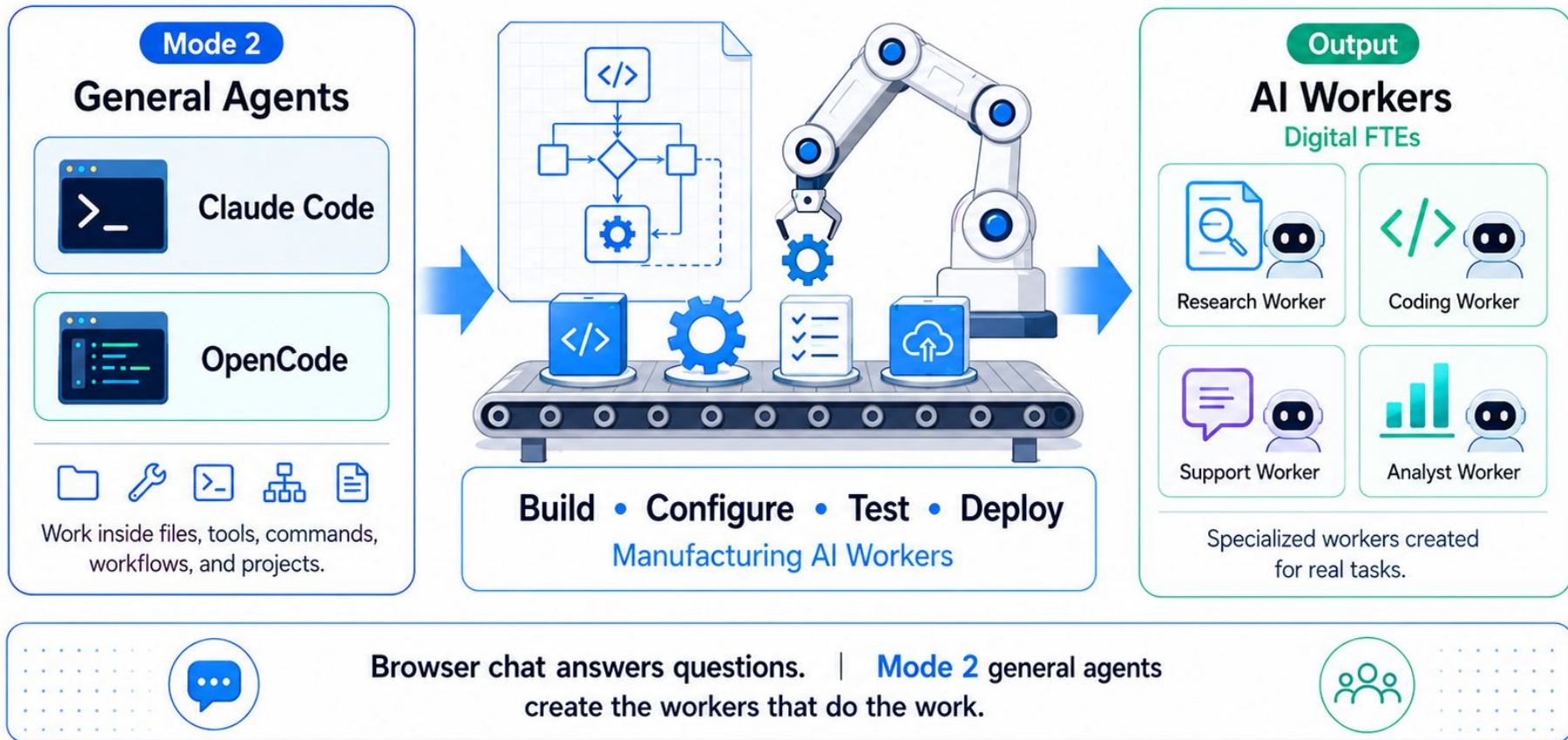
The biggest shift in AI right now isn't about making smarter tools. It's about turning AI from a tool you use into a worker you hire.

You stop paying for access to software.

You start paying for finished results.

Mode 2: Using General Agents to Manufacture AI Workers

General agents such as Claude Code and OpenCode are used to build, configure, and deploy AI Workers.



PART 06

The Agent Factory Idea



The Agent Factory

Analogy: A Garment Factory



GARMENT FACTORY

A garment factory doesn't make one shirt by accident.
It has a system.

1

Cutting



2

Stitching



3

Quality Check



4

Packing



AGENT FACTORY

The same idea — but for AI workers.

1

Design
AI Workers



2

Train Them



3

Put Them
to Work



4

Check the
Results



REPEATABLE PROCESS

Design → Train → Deploy → Verify



NOT a product

Not something
you buy.



A practice you learn

A repeatable way to build
and manage AI workers.



AI-Native Company

A company where
most workers are AI.

Agent Factory = a repeatable process for building AI workers.



The Company With Invisible Employees

Old World

200
Humans



Phones + emails



VS

New World

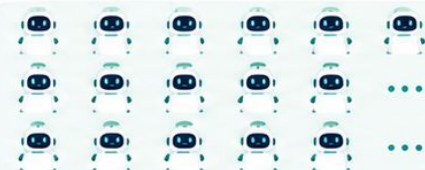
10
Humans

+

190
AI Workers



10 human supervisors
handle tricky cases
and oversee AI



190 AI workers
handle routine
questions 24/7



Routine
questions



AI handles
instantly



Complex
cases



Humans
review



Fast service
Instant answers
most of the time



Lower cost
Fewer humans
to do the work



24/7
Always on,
day and night



Most workers are AI. Humans handle the tricky cases.

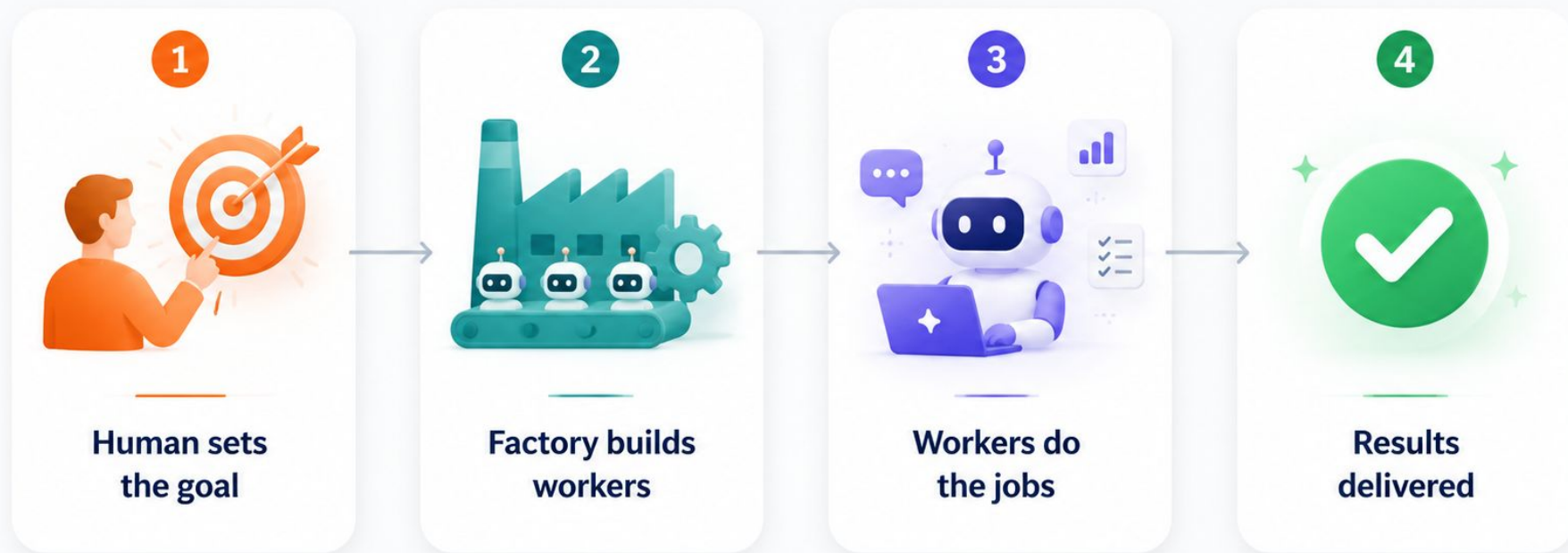
Customers barely notice the difference — they just get fast, good service.



AI-Native Company

A company where most of the
workers are AI, not human.

The Agent Factory Flow



Human gives direction → system builds AI workers → workers do the work → results return for human review

Agent Factory

The process that manufactures AI Workers and composes them into an AI-Native Company

1 General Agents



Claude Code

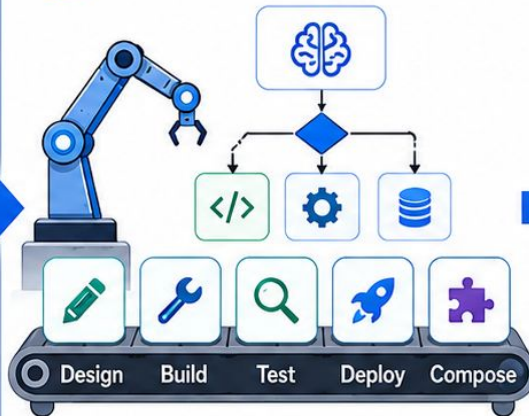


OpenCode



Used in Mode 2 to build and orchestrate AI Workers

2 Agent Factory Process



Transforms general-agent capability into specialized AI Workers

3 AI Workers



Research Worker



Coding Worker



Support Worker



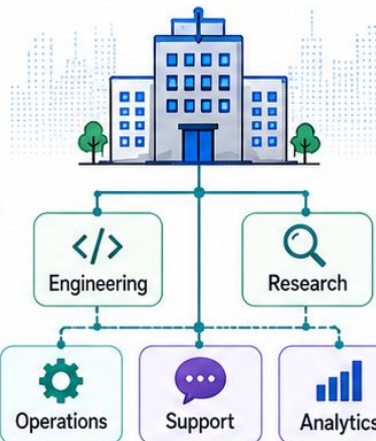
Analyst Worker



Operations Worker

Specialized Digital FTEs created for real tasks

4 AI-Native Company



AI Workers are composed into a coordinated company system



General agents create AI Workers.

Agent Factory organizes them into an AI-Native Company.



PART 07

Your Personal AI Delegate



The Executive Assistant

You set direction. Your **AI delegate** manages the work. You review the result.



Intent



Delegate



AI workforce



Approval

1 You set direction

You provide goals + guardrails

- ✓ Review Q1 report by Friday
- ✓ Budget under 5 lakh
- ✓ No meetings before 10am



CEO / YOU

2 Your AI delegate

Understands your intent



3 Delegate coordinates specialist AI workers

Behind the scenes



Secure • Private • Always on

4 You review & approve

You review the final cut



You approve the result



You do not manage
20 AI workers directly.



Your **one delegate**
manages them for you.

The Two-Layer Model

EDGE LAYER



You



Your Delegate

Your personal AI that knows your preferences, carries your authority, and manages workers.

AI WORKFORCE LAYER

**Support
Worker**

**Finance
Worker**

**Sales
Worker**

**Legal
Worker**

PART 08

The 7 Principles of General Agent Problem Solving



The Seven Principles, at a glance

Five core disciplines, wrapped by two operational principles.

P6 — CONSTRAINTS & SAFETY

scope what the agent is allowed to touch

P 1

Bash is the Key

ACTION OVER TALK

The agent has hands, not just a mouth.

Failure mode:
"It keeps narrating instead of doing."

P 2

Code as Interface

SHAPE OVER PROSE

Specify the shape before the content.

Failure mode:
"My prose request keeps getting misread."

P 3

Verification

VERIFY, DON'T TRUST

"Looks right" is the precise failure mode.

Failure mode:
"It looked right but broke in production."

P 4

Decomposition

SMALL, REVERSIBLE

Each step is cheap to undo.
Atomic units.

Failure mode:
"One big change just nuked an afternoon."

P 5

Persistence

FILES ARE MEMORY

The chat is volatile. The filesystem isn't.

Failure mode:
"It forgot what we decided yesterday."

P7 — OBSERVABILITY

see what the agent did, when, and why

Build P1–P5 inward. Wrap them with P6 (Constraints) and P7 (Observability).

PART 09

The 7 Golden Rules of an AI-Native Company (Invariants)





Building Rules That Never Change

Tools change. Structural rules don't.

✓ Non-negotiable rules



Roof
Protects
what's inside



Walls
Give structure
and shape



Foundation
Supports
everything



Choices can vary

Bricks or
concrete



Flat or
sloped roof



For AI companies:



7 rules =
your building code



Tools change
every year



Structural principles
stay the same



Tools change

Rules stay



Build on timeless rules, not temporary tools.



The 7 Golden Rules (1–4)

1 Human is always in charge

Every action traces back to a human. AI never sets its own goals.

2 Every human gets a delegate (identic AI)

You can't manage 20 workers by hand. Your delegate coordinates.

3 Workforce needs a management layer

Someone hires, assigns tasks, controls budgets, fires underperformers.

4 Each worker uses the right engine

Important work = reliable engine. Simple work = cheap engine.

The 7 Golden Rules (5–7)

5 Every worker uses a system of record

AI workers read/write the company's official memory. Without it, AI makes things up.

6 Workforce can grow under rules

When a gap appears, the system hires a new worker automatically — within human-set limits.

7 Company runs on a nervous system

Just like your body sends signals automatically, "hand touched something hot, pull back!" the AI company sends work between workers automatically. No human needed to pass tasks along. And if something breaks, the system recovers on its own.

Tools change. Rules stay.

PART 10

A Real Example: Ecomm Mart



Building the Support Worker (Steps 1–3)

Step 1

Write Clear Instructions

Define the goal: read tickets, classify them, write replies, escalate hard cases. Set budget: \$200/month max.

Step 2

Connect to Company Records

Link to Shopify orders, customer database, and ticket system so the AI can look up real data.

Step 3

Give It Skills

Teach it: how to write professional replies, how to classify tickets, when to escalate. Like a training manual.

Building the Support Worker (Steps 4–5)

Step 4

Set the Guardrails

Budget cap, approval for big refunds, audit trail of every action. The AI can never go outside these limits, all will be define in specs

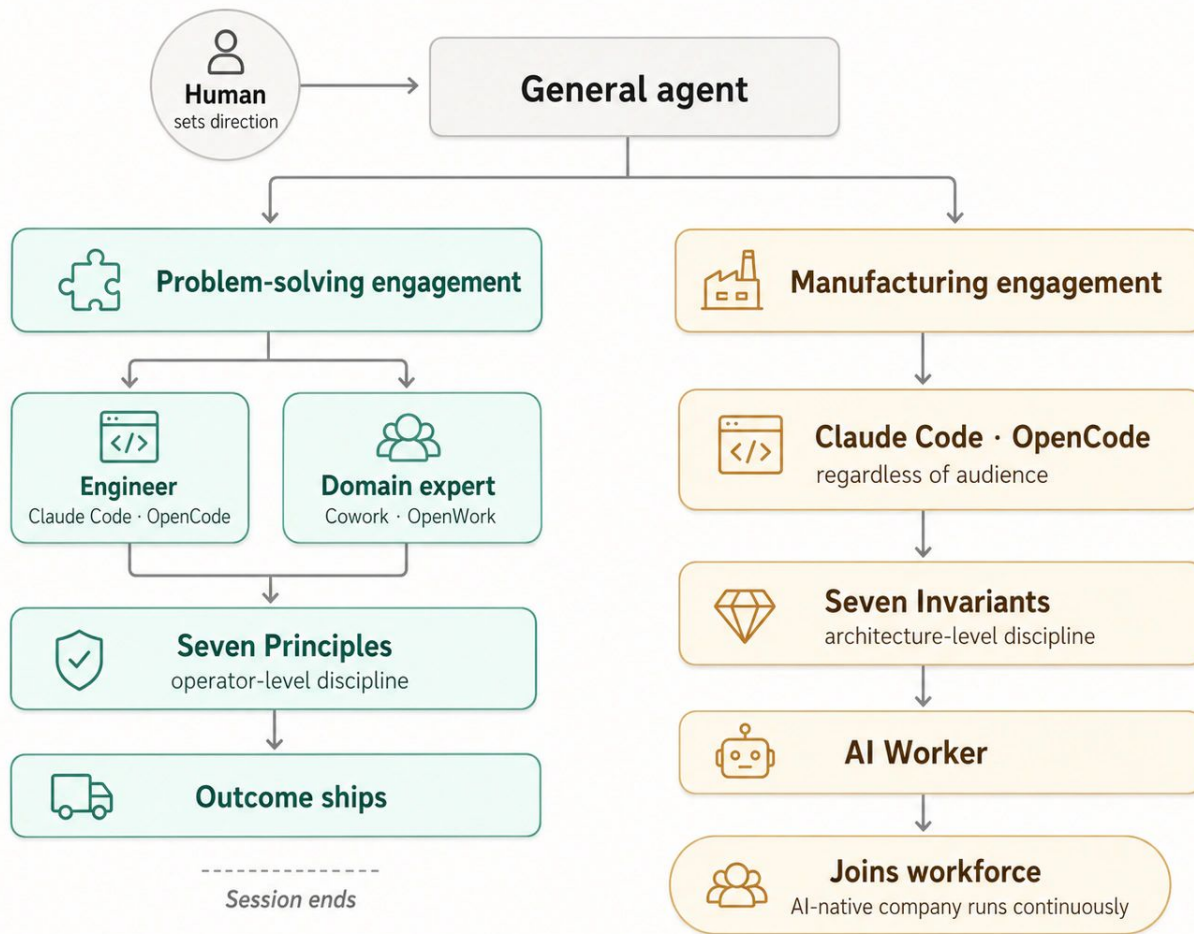
Step 5

Test, Then Deploy

Run on 100 old tickets first. Check results. Fix problems. Only then let it handle real customers.



Result: You just built a Digital FTE — a Full-Time Employee made of software. It works 24/7, stays within rules, and handles 80% of support tickets automatically.



PART 11

Key Vocabulary & What's Next



Essential Vocabulary (1/2)

AI

Software that learns from data and makes decisions

Agentic AI

AI that plans, uses tools, fixes mistakes, completes jobs

AI Worker / Digital FTE

AI built to do one specific job, like a human employee

Agent Factory

The method for building AI workers (a practice, not a product)

AI-Native Company

Company where most workers are AI, selling results not tools

Essential Vocabulary (2/2)

Delegate

Your personal AI that manages the workforce for you

System of Record

Company's official memory — the single source of truth

Spec

Clear written instruction: goal, limits, budget, success criteria

10-80-10 Rule

Human direction (10%) + AI work (80%) + Human check (10%)

TODAY'S BIG IDEAS — IN ONE BREATH

AI is software that learns and makes decisions. Agentic AI takes this further — it plans, uses tools, and completes entire jobs.

The Agent Factory is the method for building AI workers and organizing them into AI-Native Companies where AI does the work and humans set direction and check quality.

You're not being replaced — you're being promoted from worker to director.

The future belongs to people who can clearly tell AI what to do, connect it to real data, and verify the results are correct.

That's what this program will teach you. Welcome aboard.



Thank You

Your journey into Agentic AI starts now.



Based on the Agent Factory Thesis
agentfactory.panaversity.org